

## Why earthquakes happen

You may think that your feet are firmly on the ground, but Earth's crust is moving all the time. It is made of moving parts called plates. When the plates slide past or into each other, the rocks jolt and send out shock waves.

## Shaken up

The Mercalli Scale measures how much the surface of the Earth shakes during an earthquake. There are 12 intensities, or grades. At intensity 1, the effects are not felt, but by intensity 12, the shock waves can be seen and there is total destruction.



## What to do in an earthquake

Indoors, lie down under a bed or heavy table, or stand in a doorway or a corner of a room. After a minute, when the tremors will usually have finished, go outside, away from buildings, to a wide-open space.



## Earthquake words

The place within the Earth where an earthquake starts is called the focus.

The earthquake is usually strongest at the epicenter. This is the point on the Earth's surface directly above the focus. The study of earthquakes and the shock waves they send out is called seismology.

## Destructive force

A tsunami piles up and gets very tall before it crashes onto the shore. It is so powerful that it can smash harbors and towns and sweep ships inland.



Fault line

*On this side of the fault the land has moved away from you.*

A tsunami can be more than 98 feet (30 meters) high and can travel as fast as a jet.

